

The z-Transform

1 The z-Transform

Definition 1.1 (The z-Transform).

The (bilateral) z-transform of a general signal $x[n]$ is defined as

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$$

where $z = re^{j\omega}$ is a complex variable.

Remark. **Connection with the Fourier Transform**

$$\mathcal{Z}\{x[n]\} = X(z) = X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}$$

Definition 1.2 (The Inverse z-Transform).

The inverse z-transform of a function $X(z)$ is defined as

$$x[n] = \frac{1}{2\pi j} \oint_{|z|=r} X(z)z^{n-1} dz$$

A contour integration around a counterclockwise closed circular contour centered at the origin and with radius r .

2 Properties of the ROC

1. The ROC consists of circular regions centered about the origin.
2. The ROC does not include any poles.
3. If $x[n]$ is of finite duration and is absolutely integrable, then the ROC is the entire z -plane.
4. If $x[n]$ is right-sided, then the ROC is of the form $|z| > r_0$ for some r_0 .
5. If $x[n]$ is left-sided, then the ROC is of the form $0 < |z| < r_0$ for some r_0 .
6. If $x[n]$ is two-sided, then the ROC is of the form $r_1 < \mathcal{R}e(z) < r_2$ for some r_1 and r_2 .

3 Basic z-Transform Pairs

Signal	Laplace Transform	ROC
$\delta[n]$	1	$\mathbb{C}$
$\delta(n - m)$	z^{-m}	$\mathbb{C}$
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
$a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$b^{ n }$	$\frac{1}{1 - bz^{-1}} - \frac{1}{1 - b^{-1}z^{-1}}$	$ b < z < b ^{-1}$
$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\cos(\omega_0 n)u[n]$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n)u[n]$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n)u[n]$	$\frac{1 - z^{-1} r \cos \omega_0}{1 - 2z^{-1} r \cos \omega_0 + z^{-2} r^2}$	$ z > r$
$r^n \sin(\omega_0 n)u[n]$	$\frac{z^{-1} r \sin \omega_0}{1 - 2z^{-1} r \cos \omega_0 + z^{-2} r^2}$	$ z > r$

4 Properties of the Bilateral z-Transform

4.1

Signal	Transform	ROC
$x[n]$	$X(z)$	R
$x_1[n]$	$X_1(z)$	R_1
$x_2[n]$	$X_2(z)$	R_2
$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least $R_1 \cap R_2$
$x[n - n_0]$	$z^{-n_0} X(z)$	R
$x[-n]$	$X\left(\frac{1}{z}\right)$	$\frac{1}{R}$
$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$ z_0 R$
$x_{(k)}[n] = \begin{cases} x[n/k] \\ 0 \end{cases}$	$X(z^k)$	$R^{1/k}$
$x^*[n]$	$X^*(z^*)$	R
$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least $R_1 \cap R_2$
$nx[n]$	$-z \frac{d}{dz} X(z)$	R
$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - z^{-1}} X(z)$	At least $R \cap \{ z > 1\}$

4.2 Initial- and Final-Value Theorem

Theorem 4.1 (Initial-Value and Final-Value Theorem).

If $x[n] = 0$ for $n < 0$, then

$$x(0) = \lim_{z \rightarrow +\infty} X(z)$$

$$\lim_{n \rightarrow +\infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z)$$

5 The Unilateral z-Transform

5.1

Definition 5.1 (The Unilateral z-Transform).

The unilateral z-transform of a general signal $x[n]$ is defined as

$$\mathfrak{X}(z) = \sum_{n=0}^{+\infty} x[n]z^{-n}$$

where $z = re^{j\omega}$ is a complex variable.

5.2 Properties of the Unilateral z-Transform

For causal signals, the unilateral z-transform is the same as the bilateral z-transform.

5.2.1

Signal	Transform
$x[n]$	$\mathfrak{X}(z)$
$x_1[n]$	$\mathfrak{X}_1(z)$
$x_2[n]$	$\mathfrak{X}_2(z)$
$ax_1[n] + bx_2[n]$	$a\mathfrak{X}_1(z) + b\mathfrak{X}_2(z)$
$x[n-1]$	$z^{-1}\mathfrak{X}(z) + x[-1]$
$x[n+1]$	$z\mathfrak{X}(z) - zx[0]$
$z_0^n x[n]$	$\mathfrak{X}\left(\frac{z}{z_0}\right)$
$x_{(k)}[n] = \begin{cases} x[n/k] \\ 0 \end{cases}$	$\mathfrak{X}(z^k)$
$x^*[n]$	$\mathfrak{X}^*(z^*)$
$x_1[n] * x_2[n]$ both causal	$\mathfrak{X}_1(z)\mathfrak{X}_2(z)$
$nx[n]$	$-z \frac{d}{dz} \mathfrak{X}(z)$
$\sum_{k=0}^n x[k]$	$\frac{1}{1-z^{-1}} \mathfrak{X}(z)$

5.2.2 Initial- and Final-Value Theorem

Theorem 5.1 (Initial-Value and Final-Value Theorem).

For any signal $x[n]$,

$$x(0) = \lim_{z \rightarrow +\infty} \mathfrak{X}(z)$$

$$\lim_{n \rightarrow +\infty} x[n] = \lim_{z \rightarrow 1} (z-1)\mathfrak{X}(z)$$